SIMULATED
Asset Management

Investment memo (NVDA)

Agent: **Research**

OPENING PROMPT

```
What is the latest research saying about AI and cloud computing opportunities in technology companies?
```

The demo date is the latest price date in the Snowflake Public Data share at build time, and positions are month-end. Agents answer from live data, so figures, names and rankings in your environment can differ from the examples in this script.

Business Context Setup

Persona: David, Senior Research Analyst at Simulated Asset Management
Business Challenge: Investment committees require comprehensive, structured investment memos that combine quantitative financial analysis with qualitative research insights.
Value Proposition: AI-powered investment memo generation with two structured stopping points — the analyst stays in control of scope and depth.

Agent: `AM_research_copilot`
Skill: `investment-memo-generation` (7-section memo with dual stopping points), with `competitive-intelligence` and `equity-research-report` as optional branches
Data Available: SEC filing text (627k extracts from 4,124 filings, covering 104 of the 503 issuers — NVDA, AMD and INTC all included), analyst ratings with 40 vintages of history across 421 issuers, broker research, earnings transcripts, press releases

Demo Flow

Scene Setting: David needs to prepare a comprehensive investment memo for the upcoming investment committee meeting.

Step 1: Thematic Research — Where Is the Opportunity?

User Input:

PROMPT

```
What is the latest research saying about AI and cloud computing opportunities in technology companies?
```

Tools Used:

- `search_external_docs` (Cortex Search) - Search "AI cloud computing technology investment opportunities"
- `search_company_events` (Cortex Search) - Search "AI cloud strategy growth guidance"
- `search_external_docs` (Cortex Search) - Search "AI cloud product launch announcement"

Expected Response:

- AI and cloud computing investment themes from broker research — software names on AI lifting net revenue retention, semis (NVDA, QCOM, AMD) on accelerated-computing demand, GOOGL on AI-improved ad targeting with cloud as diversification
- The broker notes lean Hold, which the agent reads as the easy AI re-rating being largely complete among mega-caps

- Management commentary on AI strategy and cloud growth from earnings transcripts
- Corporate AI and cloud developments from press releases
- Synthesized technology sector opportunities with proper citations

Talking Points:

- One question searches broker research, earnings transcripts and press releases simultaneously, with citations
- The thematic view sets up the company choice: NVIDIA is the name the investment committee wants written up next

Step 2: Investment Memo — Skill Activates (STOPPING POINT 1)

Presenter Transition:

> *"The thematic research points us at the AI compute leaders. The committee wants NVIDIA written up, so let me ask for a full investment memo."*

User Input:

PROMPT

```
Generate a comprehensive investment memo for NVIDIA covering financial health, management outlook, analyst views, competitive position, and key risks. Include a Buy/Hold/Sell recommendation.
```

Skill Activation: Agent loads `investment-memo-generation` skill, executes Step 1 (Financial Foundation) using `research_analyzer` and `data_to_chart`:

- `research_analyzer` — annual financial profile (4 fiscal years), revenue by geography, and the Street view (ratings, price targets, Revenue/EBITDA/Net Income estimates). The last two follow the semantic view's `GEOGRAPHIC_REVENUE_BREAKDOWN` and `STREET_VIEW` verified queries, which work out the latest fiscal year and rating date themselves
- `data_to_chart` — two charts: **revenue by fiscal year** (bar) and **revenue by geography** (pie)

The search tools deliberately stay idle here — qualitative research is Step 2, which the analyst has not yet authorised.

Expected Response: two charts and clean tables — revenue in $B to one decimal, margins to one decimal, EPS to two. The mix chart uses geography rather than business segments: NVIDIA reports several overlapping segment hierarchies (Compute & Networking / Graphics *and* Data Center / Gaming / Automotive), which together add up to roughly 3x total revenue. NVIDIA's four regions add up to exactly 100% of revenue, which is why the chart is a pie; for a company whose regions don't cover total revenue, the agent draws a bar chart and states the coverage.

Talking Point: the skill only describes what the analyst wants — financials, geography, street view — in business terms. How to get it right in SQL lives in the semantic view: named filters (`annual_figures`, `geographic_revenue_breakdown`, `financial_estimates_usd`), verified queries, and SQL instructions for rounding and "latest period" logic, so every agent using the view benefits.

The street view in this step comes from the latest rating vintage and is Hold-leaning for the covering brokers in the memo's sample, with the share price close to the average target.

STOPPING POINT — Agent presents financial snapshot and offers:

> *"I've pulled the financial foundation. Revenue: $[X]B ([+/-X]% YoY), EPS: $[X]. I can:*
> *• **Continue to qualitative research** (SEC filings, broker research, earnings calls)*
> *• **Run earnings intelligence first** -> loads earnings-intelligence skill*
> *• **Check competitive positioning** -> loads competitive-intelligence skill*
>
> *How would you like to proceed?"*

Talking Point: The agent pauses after financials — the analyst can redirect before committing to the full memo workflow.

Step 3: Continue to Research — Four Searches (STOPPING POINT 2)

Presenter Transition:

> *"The agent paused after financials. Now we tell it to continue — it goes out to the qualitative sources, then pauses again to let us choose the memo format."*

User Input:

> **PROMPT**
>
> `Continue to qualitative research`

What Happens: The agent runs Step 2 of the skill — four Cortex Search calls, all filtered to the ticker:

- `search_sec_filings` — Risk factors, MD&A from 10-K
- `search_external_docs` (DOCUMENT_TYPE = broker_research) — Analyst views
- `search_company_events` — Management guidance from earnings calls
- `search_external_docs` (DOCUMENT_TYPE = press_releases) — Recent catalysts

Expected Response: A findings summary, not the memo:

- **Thesis direction** — Bullish / Cautious / Mixed
- **Top catalyst** (the Blackwell to Vera Rubin transition) and **top risk** — China export controls, sourced from the 10-K (the H20 charge and NVIDIA describing itself as effectively foreclosed from China's data-centre market)
- **Management tone** — Confident, on the back of data-centre growth and recovered gross margin

STOPPING POINT 2 — Agent offers:

> *"I can:*
> *• **Generate the full 7-section memo** (recommended)*
> *• **Review findings before synthesis** (raw data per section)*
> *• **Focus the memo** (e.g. risk-heavy or catalyst-focused)*
> *• **Generate an IC-ready quick thesis** (1-page, recommendation-first)*
>
> *What format works best?"*

Talking Points:

- Two stopping points, two chances to redirect — the analyst controls scope before any synthesis happens
- The findings summary cites its sources: guidance and supply commitments from the earnings calls, risk factors from the 10-K, named competitors from broker research. Every quantitative claim traces to a filing
- Four searches, each filtered to the ticker rather than scanning the whole corpus

Key Features Highlighted:

- Multi-source qualitative research across filings, broker notes, transcripts and press releases
- Broker research figures are grounded in the issuer's own SEC financials, so a note and the filing agree
- Analyst stays in control of the deliverable format

Step 4: Generate the Memo — Full 7 Sections

Presenter Transition:

> *"The research is gathered and reconciled. Now we ask for the deliverable."*

User Input:

> **PROMPT**
>
> `Generate the full 7-section memo`

What Happens: No new data retrieval — the agent synthesises everything already in the thread into the memo, calling only `data_to_chart` for a **margin trend** line chart (gross / operating / net by fiscal year) in the Financial Profile section. Mention that: it is evidence the two research steps did their job.

Expected Response: Full 7-section investment memo. In this environment the recommendation comes out as **HOLD, medium conviction**: the franchise quality justifies ownership, but the share price sits at the consensus target and the

China and product-transition risks are not being paid for.

- **Executive Summary**:
  - Buy/Hold/Sell recommendation with rationale
  - Key financial highlights (revenue growth, margins, FCF)
  - Critical risks and mitigants
  - Near-term catalysts
- **Financial Profile**:
  - Revenue mix and growth trends from SEC filings
  - Margin analysis (gross, operating, net)
  - Cash flow generation and capital allocation
  - Key financial ratios with [FACT] labels
- **Competitive Landscape**:
  - Market position from broker research
  - Competitive moat assessment
  - Direct and indirect competitors
- **Management Outlook**:
  - Forward guidance from earnings transcripts
  - Strategic priorities and capital allocation
  - Key quotes with [FACT] labels
- **Analyst Perspectives**:
  - Consensus rating and price targets
  - Bull/bear case arguments
- **Risk Assessment**:
  - Macro, regulatory, competitive risks
  - Leading indicators for each risk
- **Catalysts**:
  - Near-term events (product launches, earnings)
  - 12-24 month scenarios (bear/base/bull)

Talking Points:

- **3-turn conversation**: the opening prompt, "Continue to qualitative research", then "Generate the full 7-section memo" — two stopping points, one complete memo
- Each stopping point was a genuine fork: the analyst could have run earnings intelligence first, narrowed the scope, or asked for a 1-page thesis instead
- AI orchestrates Cortex Analyst and three Cortex Search services to build the analysis
- Structured output follows institutional investment memo standards
- Clear labelling of facts vs. analysis vs. inference for compliance
- Source citations throughout for audit trail

Key Features Highlighted:

- Multi-tool orchestration for comprehensive research synthesis
- Structured investment memo format with executive summary
- Automatic source attribution and fact/analysis labelling
- Professional output ready for investment committee

Step 5: Deep-Dive on Specific Section (Optional)

Presenter Transition:

> *"The investment memo provides a comprehensive overview, but the investment committee often wants to drill deeper on specific sections. Competitive positioning is particularly important for technology companies. Let me expand on the competitive landscape with detailed market share analysis..."*

Reasoning: Executive summaries identify key areas; deep-dives provide the supporting evidence. This demonstrates the agent's ability to refine any section without losing overall context.

User Input:

> **PROMPT**
>
> ```
> Expand on the competitive landscape section. How does NVIDIA's AI chip market position
> compare to AMD and Intel? Include market share data if available.
> ```

Tools Used:

- `search_external_docs` (Cortex Search) - Competitive analysis, market share estimates and recent developments from broker research
- `research_analyzer` (Cortex Analyst) - Revenue, margin and rating comparison across NVDA, AMD and INTC
- `data_to_chart` - Comparative trend visualisation

Note: the memo's Competitive Landscape section discusses the moat and names the rivals broker research identifies (Qualcomm, TSMC, Marvell, plus custom-ASIC threats such as Broadcom). AMD and Intel are the natural peer comparison but are not always named unprompted, so this step introduces them explicitly. Expect the agent to point out that the AMD and Intel broker notes name Qualcomm, TSMC and Marvell as their peers rather than NVIDIA — it is each house's framing, not a head-to-head share study.

Expected Response:

- Detailed competitive positioning analysis
- Market share estimates from broker research (labelled as [ANALYST ESTIMATE])
- Product comparison (GPUs, data centre, AI accelerators)
- Competitive moat assessment (CUDA ecosystem, software lock-in)
- Recent competitive developments from press releases
- Financial comparison table (revenue growth, margins) from SEC filings — NVIDIA converts far more revenue to profit than either rival
- Positioning: AMD as the credible number-two challenger and most likely share-taker; Intel a longer-dated turnaround risk; hyperscaler custom silicon flagged as the subtler threat

Talking Points:

- Drill-down capability for any section of the investment memo
- Cross-company financial comparison using SEC filing data
- Market share data clearly labelled as analyst estimates
- Competitive moat analysis synthesised from multiple sources

Key Features Highlighted:

- Section-specific deep-dive capability
- Cross-company financial analysis
- Clear data source attribution
- Competitive intelligence synthesis

Step 6: Skill-Driven Peer Set & Moat Assessment (Optional)

Presenter Transition:

> *"We've compared NVIDIA to AMD and Intel by hand. The `competitive-intelligence` skill does this as a structured workflow — it picks the peer set itself, then lets us decide where to go deeper."*

User Input:

> **PROMPT**
>
> ```
> Compare NVDA to its main competitors
> ```

Skill Activation: Agent loads `competitive-intelligence` skill:

- `research_analyzer` pulls NVDA's segment and geographic data
- `search_external_docs` finds broker research mentioning competitors
- `research_analyzer` compares financials and analyst ratings across the peer set

Expected Response: Peer set identified (AMD, INTC, AVGO) + financial comparison table + positioning verdict

STOPPING POINT — Agent offers:

> ***• Deep-dive a specific competitor***
> ***• Analyse market share trends***
> ***• Assess competitive moat***
> ***• Generate competitive positioning summary***

Follow-up — Branch into Moat Assessment:

User Input:

PROMPT

```
Assess the competitive moat
```

Expected Response: Moat assessment — **Wide** today, anchored on CUDA switching costs and full-stack integration, with durability against hyperscaler custom silicon marked as the at-risk dimension. The agent ties this back to the HOLD: the moat's long-term durability, not its current strength, drives terminal value.

Talking Point: The agent composed structured-data and unstructured-document tools to build a competitive picture — then let the analyst choose where to go deeper.

Step 7: Risk Scenario Analysis (Optional)

Presenter Transition:

> *"Strong competitive positioning is encouraging, but professional investment analysis requires equal attention to risks. Let me build a comprehensive risk framework with monitoring indicators so we can proactively track any thesis-threatening developments..."*

Reasoning: Investment committees require balanced analysis. After understanding the opportunity and competitive position, we must systematically assess what could go wrong and how we'd know.

User Input:

PROMPT

```
What are the key risks to the investment thesis? For each risk, provide leading indicators we should monitor and potential mitigants.
```

Tools Used:

- Typically none. By this point the risk material is already in the thread — 10-K risk factors from Step 3, management commentary from the earnings calls, and the competitive picture from Steps 5-6 — so the agent synthesises the framework from context rather than re-searching.
- `search_external_docs` / `search_company_events` (Cortex Search) - only if the analyst asks for a risk not already covered

Presenter note: no tool calls here is the correct outcome, not a failure. Say so — it shows the agent reusing gathered evidence instead of paying for the same search twice.

Expected Response:

- **Risk Inventory Table**:

Risk Category	Risk Description	Leading Indicators	Mitigants
Regulatory	China export restrictions	Policy announcements, trade tensions	Geographic diversification
Competitive	AMD/Intel catching up	Market share trends, product launches	R&D investment, CUDA ecosystem
Demand	AI capex slowdown	Cloud provider guidance, order trends	Diversified customer base

Risk Category	Risk Description	Leading Indicators	Mitigants
Valuation	Premium multiple compression	Interest rates, risk sentiment	Growth sustainability

- Management's risk commentary from earnings transcripts
- Analyst risk assessments from broker research
- Recent risk-related developments from press releases
- A single headline disconfirming signal: **gross margin**. The agent proposes a sustained slide below about 70% without a transition explanation as the trigger to revisit the HOLD

Talking Points:

- Structured risk framework with actionable monitoring indicators
- Multi-source risk synthesis (management, analysts, news)
- Clear mitigants for each identified risk
- Investment committee-ready risk assessment

Key Features Highlighted:

- Comprehensive risk framework generation
- Leading indicator identification for proactive monitoring
- Multi-source risk validation
- Actionable risk mitigation strategies

Step 8: Investment Committee Summary with PDF Generation (Optional)

Presenter Transition:

> *"We've completed the full analysis cycle: comprehensive memo, competitive deep-dive, and risk assessment. Now let me distil everything into a one-page brief and generate a professional PDF that the investment committee can act on immediately..."*

Reasoning: Detailed analysis must culminate in actionable recommendations. The one-page format forces prioritisation and the PDF provides a decision-ready document for the committee.

User Input:

> **PROMPT**
>
> ```
> Summarise the key points from our analysis into a one-page investment committee brief with clear recommendation and next steps. Generate a professional PDF for the investment committee.
> ```

Tools Used:

- Synthesis of previous analysis
- `pdf_generator` (Generic) - Generate branded PDF with `internal` audience

Expected Response:

- **Investment Committee Brief**:
  - **Company**: NVIDIA Corporation (NVDA)
  - **Recommendation**: HOLD, medium conviction (carried through from the memo)
  - **Price Target**: Consensus analyst target with range
  - **Key Thesis Points** (3-5 bullets):
    - AI data centre leadership and growth trajectory
    - Competitive moat via CUDA ecosystem
    - Strong financial profile with expanding margins
  - **Key Risks** (3-5 bullets):
    - Regulatory exposure to China
    - Valuation premium vs. peers
    - Customer concentration in hyperscalers
  - **Near-Term Catalysts**:

        - Next earnings release
        - Product launches
        - Industry events
    - **Recommended Action**: Specific portfolio action recommendation
    - **Follow-Up Items**: Additional analysis needed
- **PDF Generation**: Professional branded PDF with SAM logo
- **Download Link**: Presigned URL for immediate access

Talking Points:

- One-page executive summary suitable for investment committee
- Clear recommendation with supporting rationale
- Professional branded PDF ready for distribution
- Complete research workflow from analysis to formal deliverable

Key Features Highlighted:

- Executive summary generation from detailed analysis
- Clear, actionable recommendations
- Professional investment committee format with PDF output
- Complete research workflow in single session with formal deliverable

Optional Extension: Full 10-Section Equity Research Report + PDF

Presenter Transition:

> *"The memo is the committee's decision document. When the research team needs the full institutional write-up instead, the* `equity-research-report` *skill produces a 10-section report with consensus-driven bull/base/bear scenarios — and exports it as a branded PDF."*

> ***Note on Stopping Points**: The skill pauses after the financial foundation and again after research gathering. A highly specific prompt (like the one below) is read as full authorisation and produces the complete report in one response; use "Write a research report on NVIDIA" instead if you want to show the stopping points, then reply "Continue to research phase".*

User Input:

> **PROMPT**
>
> ```
> Generate a comprehensive equity research report for NVIDIA with recommendation, financial
> model, scenario analysis with bull/base/bear cases, and risk assessment.
> ```

What Happens: The agent interprets the explicit request as authorisation for the full workflow and delivers the complete 10-section report in a single response, including:

- Buy/Hold/Sell recommendation with price target
- Investment thesis with pillars table
- Financial model with historical + forecast data
- Scenario analysis (Bull/Base/Bear) with probability-weighted return
- Industry analysis and competitive landscape
- Risk assessment matrix
- Catalysts timeline
- Valuation summary
- ESG considerations
- Monitoring plan with validation/invalidation triggers

> ***Presenter note**: The equity research report can land on **BUY** with a base target above the current price, while the memo earlier in the same session landed on HOLD. The two skills weight the scenario analysis differently. If both are shown, say so rather than letting the audience spot it: the memo is the committee's conservative call, the report is the research team's scenario-weighted view.*

Follow-up — PDF Export:

> **PROMPT**
>
> `Generate a professional PDF of the complete equity research report including all sections we've discussed: recommendation, thesis, financial model, scenario analysis, industry analysis, risks, valuation, and monitoring plan.`

Tools Used:

- `research_analyzer`, `search_sec_filings`, `search_external_docs`, `search_company_events` (as for the memo)
- `pdf_generator` (Generic) - Generate branded PDF with `internal` audience

Talking Points:

- One prompt produced a complete institutional-grade equity research report
- Agent autonomously orchestrated 6+ tools (financial data, consensus, SEC filings, broker research, earnings calls, press releases)
- Scenario analysis uses real consensus estimate data (high->Bull, mean->Base, low->Bear)
- [FACT]/[ANALYSIS]/[INFERENCE] labelling throughout for compliance
- Report is immediately IC-ready
- Professional branded PDF with a presigned download link, suitable for investment committee distribution

Scenario Wrap-up

Business Impact Summary:

- **Research Efficiency**: Comprehensive investment memo generated in minutes vs. hours of manual work
- **Analysis Completeness**: Seamless integration of quantitative SEC data with qualitative research insights
- **Consistency**: Standardised memo format ensures all critical sections are covered
- **Audit Trail**: Clear source citations and fact/analysis labelling for compliance
- **Professional Deliverables**: Branded PDF documents ready for investment committee distribution
- **Coverage Capacity**: The same session can extend to a full 10-section equity research report, so analysts can cover more names

Technical Differentiators:

- **Multi-Tool Orchestration**: Automatic coordination of Cortex Analyst, Cortex Search, and PDF generation
- **Structured Output Generation**: Professional investment memo format with executive summary and PDF output
- **Source Attribution**: Clear labelling of facts, analysis, and inference with citations
- **Iterative Refinement**: Drill-down capability for any section without losing context